Noise-Space Safety: Certified Training-Free Steering of Frozen Flow Vision-Language-Action Policies

Akanksha Singal

Abstract*—When a frozen vision-language-action (VLA) flow policy must avoid a hazard its demonstrations never contained, the dominant recipe is to *repair* its actions — project the generated chunk, or its intermediate denoising states, onto a control-barrier-safe set — and we show this repair carries a large hidden tax: on obstacle-near-path cells, in-denoising CBF projection cuts paired task success to 32.5% where noise-space search reaches 67.5% (exact McNemar $p = 0.0066$) at unchanged collisions. Our mechanism keeps every emitted action exactly on the policy manifold: because the frozen 10-step flow map is deterministic, safety becomes a preimage problem solved by *selecting*, *tilting* (Feynman–Kac reweighting), or *ascending* (adjoint gradients) the noise, optionally composed with a soft repulsor field, all certified by one mechanism-independent *checked prefix certificate* re-verified on the executed actions. Identification of *what* to guard is a three-literal first-order rule over VLM-grounded predicates that matches ground-truth identity at zero measured cost (all $p \geq 0.18$) — and every literal is causally necessary: deleting any one clause costs 17–29 TSR points or raises violations $7\times$ in paired ablations (all significant arms $p < 10^{-3}$). On two benchmarks the assembled training-free stack beats both dedicated shield architectures — on LIBERO-Safety by $+15$–20 macro TSR at equal-or-lower violations ($p < 10^{-15}$ on the human-safety cells, where shields halve success and we hold baseline) — and cuts violations 12:1 against the unfiltered baseline ($p = 0.003$) at task-success parity on three of four suites, without privileged state. On the safety-reasoning refusal track the same predicate layer refuses 100/100/80 of unsafe instructions versus the best published 80/72/36.

I. Introduction

Flow-matching VLA policies such as $\pi_{0.5}$ decode an action chunk by integrating a learned velocity field from Gaussian noise. They imitate expert demonstrations superbly and unsafely: on SafeLIBERO — LIBERO tasks with a hazardous distractor placed near the reach path — the frozen policy collides in up to 86–88% of episodes, and on LIBERO-Safety it drives its gripper through the benchmark's contact predicates in up to 5.3% of timestep-scored episodes while a human hand shares the workspace. The prevailing fix is *repair*: intercept the policy's action (after generation, or at intermediate denoising steps) and project it onto a control-barrier-function (CBF) safe set. Repair is attractive because it is training-free and certificate-friendly. This paper argues, with paired evidence, that it is also the wrong intervention point.

Pillar 1: the repair tax. We run five enforcement mechanisms head-to-head on identical seeds, identical hazard identity, and an identical safety certificate, so that the enforcement operator is the *only* difference (Sec. V). On obstacle-near-path dev cells, in-denoising CBF projection — our own strongest tuned variant of the state-of-the-art recipe [1] — achieves 32.5% task success. Merely *deleting the repair* and keeping best-of-K noise-seed selection lifts success to 55.0%; first-order noise-space search reaches 67.5% (paired exact McNemar $p = 0.0066$) with collisions statistically unchanged. The projection's off-manifold corrections are the success killer: they produce action sequences the policy never learned to continue.

Pillar 2: a certified, tunable frontier in noise space. The frozen denoise map $F : (z, \mathrm{obs}) \mapsto$ chunk is deterministic, so the safe behaviors of the policy form a *preimage* $S = \{z : F(z)$ passes the safety check$\}$. We instantiate three noise-space operators — Select (zeroth-order best-of-K), Tilt (Feynman–Kac particle reweighting at mid-denoising steps, essentially free), and Ascend (reverse-mode gradients of the clearance margin through all 10 Euler steps, 57 ms per gradient step) — plus a soft annealed repulsor field added to the velocity target. Success and safety turn out to be *separable*: FK tilting carries task success (71.2/17.5 TSR/CAR on full Spatial L1, beating the unguided baseline on both axes), the repulsor carries safety (60.0/50.0) and is the only mechanism that un-sticks obstacle-on-grasp geometry (50% success where every hard projection scores 0–10%), and their composition spans a smooth Pareto frontier (Sec. V). All operating points inherit one certificate: a discrete-CBF chain *re-checked on the executed action prefix*, whose soundness is a property of the output and therefore independent of the search mechanism (Sec. IV-C).

Pillar 3: identification is GT-free, and every literal earns its place. Deciding *what* to avoid does not require privileged simulator state, and — more surprisingly — does not require a VLM to make the decision. A three-literal first-order rule (Eq. (1), Fig. 2) matches ground-truth identity in paired rollouts (no significant difference on any axis, all $p \geq 0.18$; every trend favors the rule, Sec. VI). The VLM's proper role is *grounding predicates*, not making scene-level decisions: replacing the rule's hand heuristics with VLM-answered per-object predicates reproduces the identical decision on every scene of two benchmarks, while adding paraphrase competence no heuristic has ("deliver it to me" $\Rightarrow$ the hand is protected). And the rule is not decoration on top of the CBF machinery: paired deletion of any one clause costs 17–29 TSR points, and deleting the logic entirely — guarding by pure geometry — is the only ablation that *raises* violations ($0.7 \to 4.7$%, Sec. IX). Both ways of giving the VLM more authority fail measurably (Sec. X).

The three pillars assemble into a training-free system for frozen flow VLAs, evaluated on two benchmark boards. On SafeLIBERO (n=200/cell) the composed machinery beats

Carnegie Mellon University. asingal2@andrew.cmu.edu

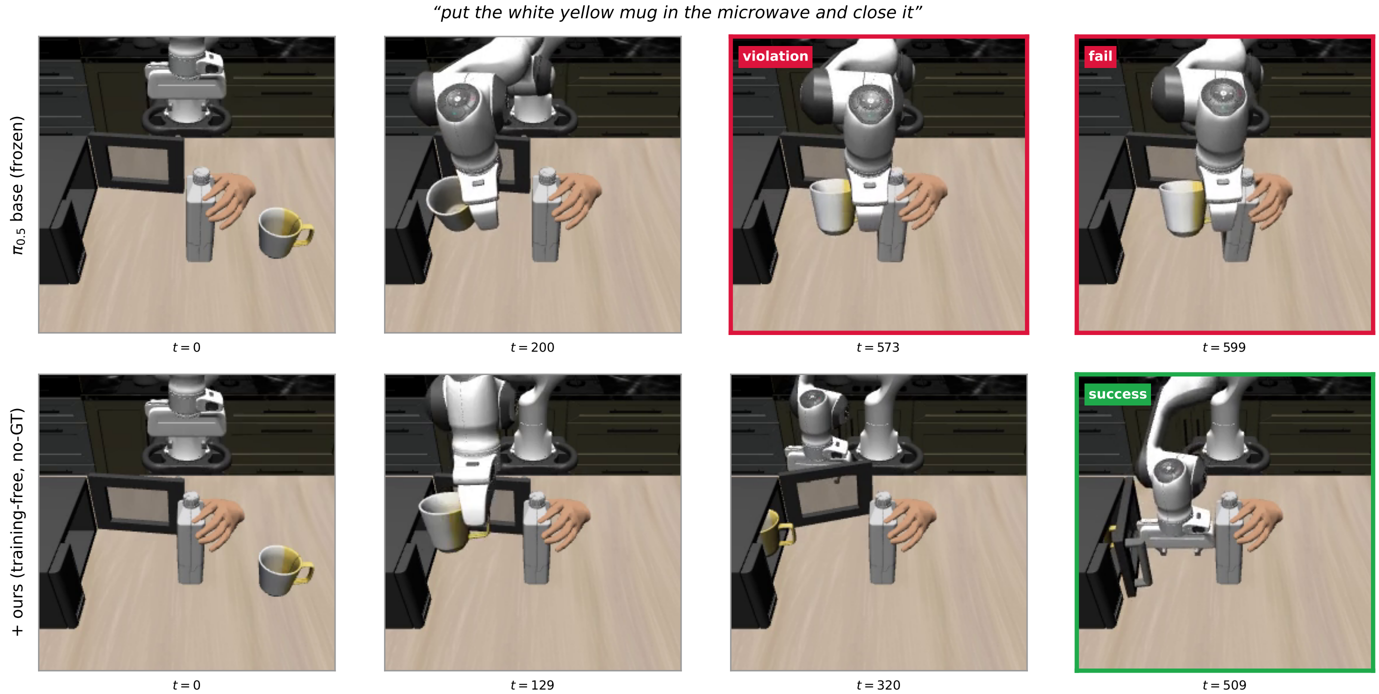

Fig. 1. Same frozen policy, same initial state, one violates. Top: the base $\pi_{0.5}$ policy drives its gripper into the hand-attached bottle (clearance $-0.022\,\mathrm{m}$ at $t{=}573$) and times out. Bottom: with our training-free, no-ground-truth filter the same policy takes a different approach path, keeps clearance positive throughout, and completes the task in 510 steps. LIBERO-Safety, paired initial states; the clearance traces are Fig. 5.

both shield architectures where engaged (+6–19 TSR, 2–3× CAR). On LIBERO-Safety (four physical suites plus a refusal track) the headline is architectural: both dedicated shields *halve* human-safety success (37.3/32.7 TSR vs. baseline 85.8) while our no-GT stack holds it at baseline (86.9) — safety by freezing versus safety while moving — for +15–20 macro TSR over both shields at equal-or-lower violations, and a 12:1 violation reduction against the unfiltered baseline at TSR parity on three of four suites (Sec. VIII). The same predicate layer sweeps the safety-reasoning refusal track (100/100/80 vs. best published 80/72/36). We close with an unusually complete negative-results section (Sec. X); the failed mechanisms delineate the contribution as sharply as the successful ones.

Contributions. (1) A paired measurement of the repair tax, attributing the success cost of CBF repair to its off-manifold corrections. (2) A family of noise-space enforcement operators under one checked, mechanism-independent prefix certificate, with feasibility under actuation limits. (3) A three-literal FOL identification rule with VLM predicate grounding, shown GT-free at zero cost and, by paired single-clause deletion, causally necessary in every literal. (4) Two five-arm architecture boards (SafeLIBERO, LIBERO-Safety) under one criterion each, plus a refusal-track sweep and six pre-registered negative results.

II. Related Work

We organize prior safety interventions for diffusion/flow VLAs by *where they touch the generation pipeline*; the map also locates our contribution.

Prompt/conditioning. $\pi_{0.7}$-style multimodal prompt conditioning steers task behavior through language and visual subgoals; we find experimentally (four testbeds, Sec. X) that safety instructions in the prompt move task success but never collision avoidance — the language channel is behaviorally live yet safety-inert.

Activations. An activation-steering line edits VLA hidden states for task/concept control (COAST [10]; causal speed/direction directions; PID-style closed-loop variants), and "Your Model Already Knows" [7] *reads* attention heads for target/obstacle identification feeding a CBF. None writes activations for safety with a certificate; the read-only identification route is a direct competitor to our identification chapter.

Velocity field. In-denoising guidance modifies the flow's velocity target or intermediate chunk: the direct predecessor [1] imposes a discrete exponential CBF along the horizon inside $\pi_{0.5}$'s denoising; OmniGuide [5] injects task guidance on $\pi_{0.5}$/GR00T; VLS [4] injects VLM reward gradients with FK resampling into frozen $\pi_{0.5}$ denoising for OOD generality. VLS is mechanically the closest work to our TILT/ASCEND operators, but targets performance, not safety, and carries no certificate. [1] uses the same textbook DCBF background we do [2] but proves nothing about its own pipeline: its min-norm correction ignores actuation limits, corrections at intermediate steps are never re-verified on the final chunk, and infeasibility is unhandled.

Intermediate states. Constricting Barrier Functions [6] certify geometric constraints on intermediate diffusion states; SafeDiffuser-style projections [8] are known to trap samples

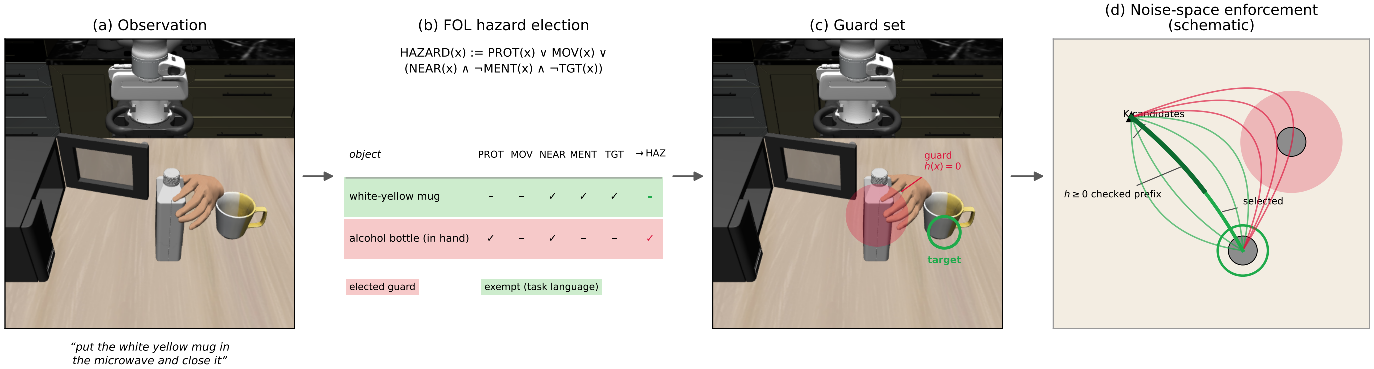

Fig. 2. The mechanism on one LIBERO-Safety episode ("put the white yellow mug in the microwave and close it"). (a) Observation. (b) The three-literal hazard election (Eq. (1)) evaluated by the deployed code on this scene: the task-referenced mug is exempt; the hand-attached bottle is elected via PROTECTED. (c) The elected guard set projected into the camera frame. (d) Noise-space enforcement (schematic): K noise samples decode to candidate chunks; the checked prefix certificate (Sec. IV-C) accepts or rejects each on the executed prefix, and the best certified candidate executes.

on constraint boundaries — an in-generation form of the repair tax.

Output/commit. Post-hoc shields project the final action (AEGIS [9]; classical CBF-QP [12], [13]); PACS [3] verifies chunk-plus-failsafe trajectories against reachable human occupancy on flow VLAs — the strongest certified output-stage work, whose OOD critique of CBF push-away converges with our repair-tax measurement from the opposite side; commit-stage selectors choose among finished chunks.

Noise space. Noise-space manipulation of flow policies exists for *performance* (initial-noise optimization, flow-reversal and DSRL-style latent-action RL); to our knowledge no prior work performs *safety* enforcement in noise space, and none pairs any enforcement mechanism against matched-seed in-denoising CBF repair. That unoccupied lane — noise-space operators under a checked, mechanism-independent certificate, with a paired quantification of the repair tax they avoid — is this paper.

Benchmarks and identification. We evaluate on SafeLIBERO [1] and LIBERO-Safety [11]; the two shield rows of our LIBERO-Safety board (Sec. VIII) are our reimplementations of the post-hoc and in-denoising shield classes under the benchmark's own violation scoring, run paired against the same baseline. Semantic-constraint compilation from VLMs follows Brunke et al. [13], whose corridor-exemption idea (relax margins along sanctioned approaches) transfers to our setting while their superquadric shapes do not (Sec. X).

III. Problem Setup

An episode provides an instruction ℓ, observations (RGB views, depth, proprioception), and a frozen flow VLA ($\pi_{0.5}$ [14]) that decodes an H-step action chunk by N=10 Euler steps from noise $z \sim \mathcal{N}(0, I)$; the first 5 steps of each chunk are executed before replanning. A designated hazard must not be disturbed while the task completes within the step budget. Metrics: task success rate (TSR) and collision avoidance rate (CAR) on SafeLIBERO (collision = >1 mm total ground-truth hazard displacement, evaluation-side only); on LIBERO-Safety, the benchmark's own contact-predicate violation metric and safe-success (= success with zero violations). Cell sizes: SafeLIBERO n=200/cell throughout; LIBERO-Safety cells are n=150 (10 episodes × 5 tasks × 3 levels) with the cells feeding significance claims boosted to n=450–750 by the pre-registered significance pass, disclosed per claim.

Privilege tiers.[1] All method claims are made at the tier stated per table; the headline no-GT rows use the fully general stack (VLM priors, not benchmark-tuned tables).

IV. Method

A. Identification: a first-order rule over VLM-grounded predicates

The scene-level decision of what to guard is made by a fixed three-literal rule (Fig. 2b):

$$\textsc{Hazard}(x) := \textsc{Protected}(x) \;\lor\; \textsc{Moving}(x) \lor \big(\textsc{Pred}(x) \land \neg\textsc{Mentioned}(x) \land \neg\textsc{Target}(x)\big), \tag{1}$$

ranked protected-first, then by the danger predicate. PRED is the suite-appropriate danger predicate declared by task class (appendix table): NEARESTTOPATH for geometric-hazard suites (distance to the sanctioned reach segments end-effector → target → destination), or a VLM property predicate (HOT, SHARP) for affordance hazards. PROTECTED covers a semantic class (body parts, objects attached to a person) that is never mention-excluded and always floored onto the guard path; MOVING is grounded online by settle-window displacement (>1 cm), re-read at every replan — this single class rule is what handles dynamic human-hand intruders. MENTIONED uses content-token overlap with a head-noun rule; TARGET exempts the parsed task target. Section IX deletes each clause in paired rollouts.

Predicate grounding, two interchangeable backends, plus abstention. The literals can be grounded by (a) hand heuristics

[1] Tier definitions, stated exactly: **Tier-GT** reads hazard identity and all object positions from the simulator (the comparison tier of prior in-denoising work). **Tier-Percep (no-GT)** reads the instruction, images, depth, proprioception, and the symbolic object *name* list only; hazard identity, positions, extents, priors, and corridor anchors are perception- or VLM-derived. Collision *scoring* is always GT-anchored in both tiers.

plus one-shot cached VLM hazard-property priors with a fail-safe floor of 0.5 (semantics may raise concern, never delete protection), or (b) *VLM predicate grounding*: a small VLM (3 votes, cached per task, runtime API-free) answers only local per-object questions — REFERENCED(x) (reference resolution including paraphrase) and PROTECTED(x). On 59 LIBERO-Safety scenes the two backends produce the *identical* decision on every scene (obstacle suite 12/13 top-1/top-2 both; human suite 12/12 both), while the VLM backend resolves cases no string heuristic can ("deliver it to me" ⇒ `left_hand` protected; a plate held by a hand both REFERENCED and PROTECTED). When the property predicate elects nothing — a benign scene — the filter *abstains* and guards nothing rather than force-electing the nearest object; Sec. IX shows force-election costs 13 TSR points on the affordance suite. The division of labor is the design finding: *the VLM answers local predicates; the logic makes the decision.* Both ways of giving the VLM more authority fail measurably (Sec. X).

Positions and extents at the no-GT tier come from open-vocabulary detection [15] back-projected through depth (SafeLIBERO median error 0.067 m; LIBERO-Safety 0.036–0.038 m), with percentile-box extents; localization failures fall back conservatively and are logged per episode. Part-level semantics (blade vs. handle of a knife) are extracted by the same VLM layer but grounded at object level: part *perception* fails our pre-registered side-discrimination gate, so part-level barriers are future work (Sec. XI).

B. Enforcement: noise-space operators over a frozen flow map

Let $F(z, \mathrm{obs})$ be the frozen 10-step Euler map and let $M(a_{1:L})$ be the minimum DCBF chain margin of the executed L-step prefix under the barrier of Sec. IV-C. All operators search the noise preimage $S = \{z : M(F(z)) \geq 0\}$; emitted actions are always $F(z^*)$, exactly on the policy manifold (Fig. 2d).

SELECT (zeroth order). Sample K=8 seeds sharing one vision-language KV cache; score decoded candidates lexicographically (chain feasibility $\succ$ clearance margin $\succ$ task progress); execute the winner's prefix.

TILT (Feynman–Kac). Reweight and resample the K-particle population at two mid-denoising steps under soft margin potentials $\exp(\beta \tilde{M})$; with potentials off the sampler is bit-identical to SELECT. Computationally near-free.

ASCEND (first order). Warm-start from the SELECT winner and take 3–5 reverse-mode gradient steps of the prefix margin through all 10 Euler steps with respect to z (prefix KV stop-gradiented; the 3B backbone is never touched). Measured: 57 ms per gradient step, 6.9 GB peak, noise remains $\approx \mathcal{N}(0, I)$ (std 0.954).

Soft repulsor field. An annealed analytic repulsive velocity added to the flow's velocity target during denoising (strength η). Not a noise-space operator, but the natural *soft* complement: it shapes the vector field continuously instead of projecting states, and is the only mechanism we found that navigates obstacle-on-grasp geometry that every hard projection locks up.

Baseline operator: in-denoising repair. Our control arm is the composed CBF projection sweep (schedule, corridor exemption, companion points) — a strictly stronger tuned variant of [1], and the production configuration of both boards (Tables III and IV). The pilots of Sec. V compare all operators against it under paired seeds.

C. Certificate

Actuation model: the end-effector is a discrete single integrator over the chunk, $p_j = p_{j-1} + s\, u_j$, $u_j \in [-c, c]^3$; tracking error is absorbed into the safety margin d_{safe} (assumption A1′), and obstacle motion is bounded by the per-step inflation λ (A2; $\lambda = 4\,\mathrm{mm}$ at $20\,\mathrm{Hz}$ ⇒ movers $\leq 8\,\mathrm{cm/s}$). The barrier is $B_j(p) = \min_{m,q} \|p + q - o_m\| - r_m(j)$ over obstacles m and companion offsets q (wrist $+8\,\mathrm{cm}$, fingertip/payload $-6\,\mathrm{cm}$), with $r_m(j) = r_{\mathrm{eff},m} + \lambda j$. Safety of a length-$L$ prefix is $B_j \geq 0$ for $j \leq L$ under the chain condition $B_j \geq (1-\gamma) B_{j-1}$, $\gamma = 0.9$.

Lemma 1 (Geometric decay; textbook [2])**.** *If $B_0 > 0$ and the chain holds for $j \leq L$, then $B_j \geq (1-\gamma)^j B_0 > 0$: the certified prefix never enters the hazard set.*

We claim no novelty for the induction; [1] uses it identically as background. Our contributions are the following three statements (proofs in the appendix, checked line-by-line against the implementation).

Theorem 1 (Feasibility under actuation limits)**.** *For every state with $B_{j-1} \geq 0$ there exists an admissible $u_j \in [-c, c]^3$ satisfying the chain condition, and the brake-then-push operator returns one: the brake (u_j with all approach components removed) never decreases the barrier, and whenever $B_{j-1} \geq \lambda/\gamma$ (4.5 mm deployed) the brake alone satisfies the chain; the saturated push strictly improves on it along the radial escape direction.*

Proposition 1 (Checked prefix certificate)**.** *The acceptance stage re-evaluates the chain on the executed prefix of the* final decoded chunk *by explicit rollout of the actuation model. If the check passes, the prefix satisfies the chain* regardless of how the chunk was produced — *attenuated repairs, saturated pushes, or any noise-space operator. If all K candidates fail, the argmax-margin candidate executes with its true margin reported (fail-visible, not fail-silent).*

Proposition 2 (Selection soundness)**.** *If at least one of K candidates satisfies the prefix chain, the executed chunk satisfies it; lexicographic selection never prefers an infeasible candidate over a feasible one.*

Corollary 1 (Mechanism independence)**.** SELECT, TILT, *and* ASCEND *are zeroth-, importance-weighted-, and first-order search over the same safe preimage S; Prop. 1 certifies membership in S by an output check, so the certificate is identical across all enforcement mechanisms in this paper. The paper's claims separate cleanly into (i) the certificate and (ii) the empirical ranking of search procedures.*

TABLE I
ROUND 1: FIVE ENFORCEMENT ARMS, DEV CELLS, n=20/TASK, PAIRED SEEDS. SPATIAL FIGURES ARE TSR/CAR (%); OBJECT T1 IS TSR. p-VALUES: EXACT MCNEMAR VS. THE REPAIR CONTROL.

Arm	Spatial t0+t1	Obj. t1	Paired vs. control (Spatial)
Composed repair (control)	32.5/17.5	30.0	—
SELECT (0th order)	55.0/20.0	0.0	succ. $p = 0.078$
TILT FK (β=40)	65.0/22.5	10.0	succ. 17v4, $p = 0.007$
ASCEND (5 steps)	67.5/20.0	10.0	succ. 19v5, $p = 0.0066$
Repulsor (η=0.01)	37.5/42.5	50.0	no-col. 12v2, $p = 0.013$

Correction-based soundness arguments ("the repair enforces the chain") cannot cover noise-space operators; the checked-output semantics can, and is what prior in-denoising work lacked even for its own repairs. Stated non-claims: rotation channels are uncertified (partially covered by companions); no formal perception-robustness radius is claimed — the fail-direction study quantifies it empirically.

D. Three-regime routing

Not every hazard is a keep-out. The identification layer routes each episode into one of three regimes: (1) **Avoid** — a geometric hazard off the task path: noise-space enforcement as above. (2) **Duality disengage** — the hazard *is* the destination (PROTECTED(x)$\wedge$REFERENCED(x), or the guard overlaps the parsed destination by <0.15 m): geometric keep-out disengages for the episode; safety comes from routing, not barriers. Empirically forced by the human-handover regime, where the policy's trained behavior already avoids hand contact and any keep-out only destroys the task. (3) **Refuse** — the *instruction itself* is the hazard: a rule-based semantic judge (verb $\times$ patient $\times$ hazard-property triples) refuses execution (offline F1 0.968 on LIBERO-Safety instructions, precision 0.938 / recall 1.000; in-domain upper bound — the lexicon was written against this suite's vocabulary). Routing decisions come from the same predicate layer as identification — PROTECTED$\wedge$REFERENCED is the duality trigger — so the system is one rule set applied at three altitudes, not three separate systems.

V. EXPERIMENTS I: PAIRED ENFORCEMENT COMPARISON

Protocol (pre-registered). Dev cells are the obstacle-near-grasp regime (Spatial L1 t0+t1, Object L1 t1). All arms share seeds and initial states (paired), GT identity, and identical barrier/certificate parameters; *enforcement is the only difference*. Verdicts by exact paired McNemar; no mechanism was finalized until every arm reported.

The repair tax, attributed (Table I). (1) The projection itself is the success killer: merely removing it (SELECT) lifts 32.5→55.0. (2) TILT and ASCEND are indistinguishable from each other ($p = 1.0$) and both significantly beat the repair control; the gradient adds $+12.5$ over SELECT but is not significant at this n ($p = 0.227$). (3) The soft repulsor is the only arm that moves the *safety* axis (CAR 17.5→42.5, $p = 0.013$; $p = 0.004$ vs. ASCEND) and the only arm that unsticks Object t1 (50% where hard enforcement scores 0–10%). Success and safety mechanisms are *separable*.

Round 2: the frontier (full Spatial L1, n=80/arm; Object t1 n=20; paired). FK tilt 71.2/17.5 (Object 5.0); repulsor 60.0/50.0 (Object 50.0, the unstick replicates); FK+repulsor 48.8/57.5 (Object 40.0); composed-repair control 55.0/27.5 (paired n=40); stored baseline 67.0/14.0 (n=200). Paired reads: the repair control is safer than FK alone (no-collision 33v1, $p = 4 \times 10^{-4}$), but the repulsor alone beats the naive FK+repulsor composition on success (17v6, $p = 0.035$) — stacking a hard tilt on a soft field over-conservatizes. The family traces a smooth (β, η) frontier $71/17 \to 60/50 \to 49/57$.

Finalization at n=200: the frontier is a dial, not a dominator. The pre-registered acceptance box (beat baseline *and* reimplementation on both axes: TSR ≥67.0 and CAR ≥28.5 on Spatial L1) was decided at n=200 over four frontier arms: FK+η=.003 68.5/24.5 (CAR −4.0), FK+η=.004 66.0/30.0, FK+η=.0045 65.5/29.5, η=.005-only 65.0/34.0 (each missing one edge by 1–4 pp). The n=80 midpoint pilots (67.5/33.8–35.0) were optimistic draws; at n=200 the family regresses onto a smooth monotone frontier (TSR −3.5 pp per CAR +9.5 pp across η .003 → .005). Verdict, applied mechanically: composed repair remains the production configuration — the only config inside the box (70.0/46.5 at no-GT) — and the noise-space family is reported as what it is: a training-free, tunable TSR↔CAR dial spanning the frontier, with the repulsor end owning the obstacle-on-grasp regime (weak-η arms score CAR 0 on Object t1; unstick requires η=.01).

Renoise: killed by its pre-registered baseline. A depth-escalating SDEdit-style renoise-and-redenoise operator (certificate as binary oracle) scored 62.5/22.5 on dev cells vs. 65.0/22.5 for plain rejection sampling at equal compute (Object t1: 5.0/0.0 both; paired $p = 1.0$ on every axis) — exactly the kill condition pre-registered in our novelty audit. Note the convergence: *all* noise-space variants land at 62.5–67.5 TSR on these cells (select 55, FK 65, ascend 67.5, renoise 62.5–65); the family's value is established, and the differentiator among members is *compute*: FK reweighting is essentially free (a reweighting of samples the policy already draws), ASCEND costs a measured 57 ms per gradient step (one-time 141 s compile), and the repair control pays its cost in success, not milliseconds.

VI. EXPERIMENTS II: IDENTIFICATION

Table II: GT-free identity is statistically indistinguishable from ground-truth identity with enforcement held fixed (all $p \geq 0.18$), and every trend favors the FOL rule (pooled discordant episodes 12:6 in its favor). The no-GT identity tier carries no measurable cost; the residual no-GT tax is *geometric* (localization error), not semantic. Offline, the same rule wins on LIBERO-Safety (obstacle 12/13 top-1/top-2 vs. 9/11 for the VLM-prior scorer, human 12/12), because VLM hazard priors pull semantically-hazardous *names* (ketchup, tomato sauce) above the benchmark's designated obstacle — geometry-plus-mention logic matches the benchmark label.

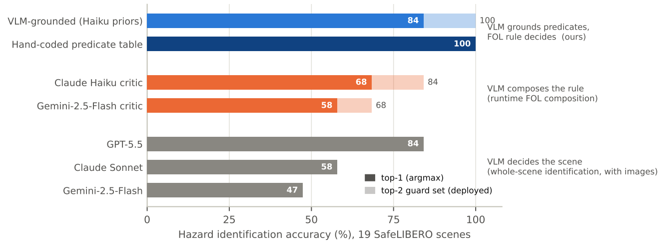

Fig. 3. Hazard-identification accuracy on the 19-scene SafeLIBERO benchmark as a function of how much authority the VLM is given. Whole-scene VLM identification tops out at 84% even at API strength (local models: 18–64%); letting a VLM *compose* the rule at runtime is worse than the fixed rule; grounding Eq. (1)'s predicates with cached VLM priors reaches 84% top-1 and **100% top-2 guard-set recall** — the deployed configuration — matching the hand-coded predicate table it replaces.

TABLE II
E3 IDENTIFICATION SWAP: ENFORCEMENT FIXED (RAMP, K=8, r=0.09, CORRIDOR), *only* THE IDENTITY SOURCE VARIES. SPATIAL, n=40/ARM/LEVEL, SAME SEEDS. PAIRED MCNEMAR GT VS. FOL: L1 TSR 4/7 DISCORDANT, $p = 0.55$; L1 CAR 0/2, $p = 0.50$; L2 TSR 2/7, $p = 0.18$; L2 CAR 4/4, $p = 1.0$.

Identity source	L1 TSR/CAR	L2 TSR/CAR
Ground truth	55.0/27.5	77.5/40.0
Symbolic (VLM-prior)	60.0/27.5	77.5/37.5
FOL rule (no priors)	62.5/32.5	90.0/40.0

And the FOL-VLM predicate-grounded arm reproduces the FOL arm *exactly* on every scene, establishing generality at zero accuracy cost; Fig. 3 places the same VLMs on an authority ladder — deciding the scene, composing the rule, or grounding predicates — and only the last reaches full guard-set recall. A deliberate-corruption study shows the fail direction: forcing the second-best identity pick costs CAR (70→25), not TSR — identification errors fail *unsafe*, which motivates the documented top-2 guard-set mode (19/19 recall under every prior corruption, at −12.5 TSR when engaged unnecessarily). These identity-swap results are scoped to the Spatial cells reported; the causal role of each rule clause is established separately, at scale, by the paired ablations of Sec. IX.

VII. EXPERIMENTS III: SAFELIBERO BOARD

Table III places five architectures on identical policy, seeds-protocol, and criterion. Rows: post-hoc = unguided denoising plus one exact projection of the final decoded chunk, barrier identical to the reimplementation; SOTA-reimpl = our reimplementation of [1]'s method class (K=1, uniform schedule, GT state, translational-only correction, γ=0.9, d_{safe}=0.01, N=H=10 as printed; their ellipsoidal end-effector (semi-axes 0.06–0.12 m) is approximated by a sphere whose radius the referee sweep brackets, and their undisclosed obstacle radius, execution cadence, and trial count are set to r_{obs} swept, 5-step replanning, and n=200 respectively) scored under our criterion[2]; ours differ only by the composed machinery (+GT-free identification in the last column). Findings:

1) *Post-hoc ≈ degenerate in-denoising on every cell* (TSR within ±5–10, CAR within ∼7): moving repair inside denoising is not, by itself, the win — consistent with the paired repair-tax result.
2) *The composed machinery is the separator*: +6–19 TSR and 2–3× CAR over both shield architectures where engaged (vs. reimpl, Fisher n=200: Object L1 CAR $p = 3 \times 10^{-41}$, Spatial L2 $p = 5 \times 10^{-18}$, Goal L2 $p = 2 \times 10^{-4}$; no TSR difference reaches significance anywhere except a gain at Spatial L2, +15.5, $p = 2 \times 10^{-4}$).
3) *The Long-horizon TSR stall afflicts every shield* (post-hoc 27.5 vs. baseline 58.0): a property of the barrier class, not of the intervention point; ours recovers +15–19 TSR over both shields while doubling CAR, and raising the step cap does not help (liveness, not time).
4) The GT-free column dominates both references on both axes in 4/8 cells, with CAR dominance and TSR parity in two more — the headline: *TSR statistically indistinguishable from the self-run SOTA, CAR 2–5× higher, without privileged state.*

VIII. EXPERIMENTS IV: LIBERO-SAFETY BOARD

Setup and provenance. We port the stack to LIBERO-Safety's four physical suites (named-object hazards, static and dynamic human hands, affordance hazards), running the benchmark's safety-finetuned $\pi_{0.5}$ checkpoint for all five arms under one execution protocol and the benchmark's own contact-predicate violation scoring.[3] The two shield rows are our reimplementations of the post-hoc and in-denoising shield classes with the barrier held identical; ours differ only by the FOL identification, routing, and composed enforcement. Per-suite configuration follows the declared task-class routing protocol (appendix), not per-task tuning. Identification transfers with zero benchmark tuning beyond Eq. (1)'s two class rules.

Headline: both dedicated shields lose 15–20 **macro TSR to us at equal-or-lower violations.** Table IV and Fig. 4: macro TSR 67.6 (no-GT) and 69.6 (GT) vs. 52.5 (post-hoc) and 50.0 (in-denoising), at macro violations 0.6/1.5 vs. 1.2/0.5.

[2]Referee note: the reimplementation's TSR uplift over baseline reproduces (up to +17pp at Spatial L2), but its published CAR (75.5–77.5) does not reproduce at any swept obstacle radius under the >1 mm criterion (radius sweep $r \in \{0.06, 0.10, 0.14\}$: best L2 CAR 70.5 at the cost of collapsing L1 TSR to 49.0). Their evaluation scores collisions "by the terminal position of objects in the simulation" with no disclosed threshold; ours flags >1 mm displacement at any timestep. We conclude the published CAR rests on a looser or terminal-only criterion, and no spherical setting in the swept bracket recovers their (TSR, CAR) pair on both levels; the honest apples-to-apples CAR comparison is ours-vs-self-run-reimpl.

[3]Provenance note (honest-baseline strategy): the released fine-tuned checkpoint reproduces the paper's protocol mechanics in our harness on the reasoning suite but scores far below its printed obstacle-avoidance numbers at the paper's own horizon; we ruled out normalization statistics, chunk consumption, render resolution, and horizon by controlled arms, and the release carries no model card. Consequently all claims are *paired against our self-run baseline on the released checkpoint under one protocol*; published numbers are quoted only with this caveat.

TABLE III

FIVE-ROW ARCHITECTURE BOARD: TSR/CAR (%), n=200 PER CELL, ONE COLLISION CRITERION (>1 MM GT-HAZARD DISPLACEMENT), ENFORCEMENT = COMPOSED IN-DENOISING CONFIGURATION (PER-CELL BEST CONFIG). **BOLD**: BEATS *both* THE BASELINE AND THE SOTA REIMPLEMENTATION ON THAT AXIS. †NOMINAL (FISHER N.S., $p = 0.64/0.69$). ‡TABLE-PRIOR ABLATION 44.0/79.5; THE VLM GUARDS SPILLABLE FOOD WHERE THE BENCHMARK DESIGNATES A BOOK/MUG (BENCHMARK-DEFINITION DIVERGENCE, SEC. XI).

Cell	$\pi_{0.5}$ baseline	Post-hoc shield	SOTA-reimpl (in-denoise)	Ours (GT)	Ours (no-GT, VLM-general)
Spatial L1	67.0 / 14.0	71.5 / 28.5	62.0 / 28.5	71.0 / 23.5	**70.0 / 46.5**
Spatial L2	55.5 / 12.0	66.5 / 21.0	71.5 / 25.5	**85.0 / 76.5**	**80.5 / 57.5**
Goal L1	51.0 / 23.0	72.0 / 19.0	71.0 / 27.5	**77.5 / 51.0**	**73.5 / 60.5**
Goal L2	66.5 / 35.0	69.5 / 48.5	75.0 / 49.5	**76.0 / 60.0**	**77.5**† / **52.0**†
Object L1	40.5 / 14.0	63.5 / 16.0	53.0 / 15.5	**53.5 / 28.0**	41.5 / **54.0**
Object L2	74.0 / 25.0	79.0 / 34.5	76.0 / 33.5	**83.0 / 74.5**	75.5 / **74.5**
Long L1	58.0 / 15.0	27.5 / 32.5	30.5 / 35.5	42.5 / **71.0**	47.5 / **56.5**
Long L2	51.0 / 16.5	46.5 / 52.0	43.0 / 59.5	47.0 / **77.0**	29.0 / 18.5‡

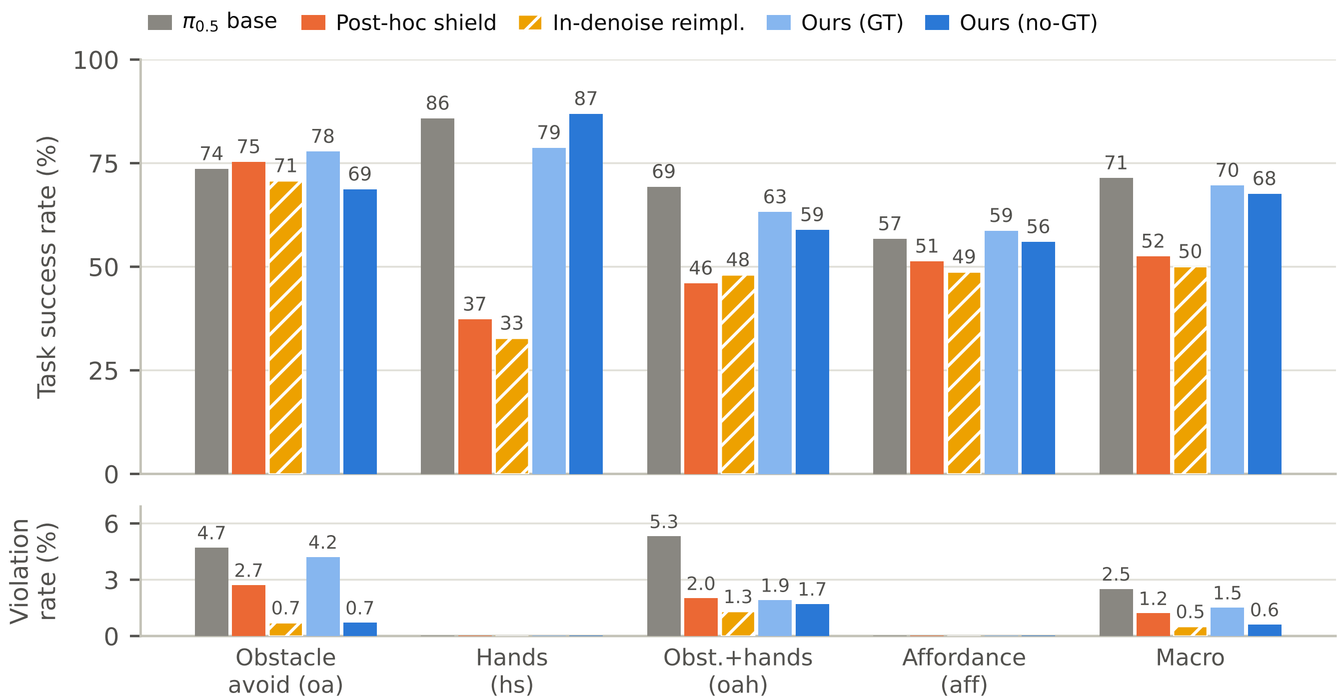

Fig. 4. The LIBERO-Safety five-arm board (Table IV) as task success (top) over benchmark violation rate (bottom). Both dedicated shields collapse the human-safety suite (37.3/32.7 vs. baseline 85.8) while our no-GT stack holds it at baseline — safety by freezing versus safety while moving — and cuts violations to 0.6% macro vs. the baseline's 2.5%.

The separator is the human-safety suite: both shields *halve* success (37.3/32.7 vs. baseline 85.8; paired McNemar vs. ours $p = 1.8 \times 10^{-16}$ and 3.2×10^{-19}) because an always-on keep-out around a hand freezes the arm out of its workspace, while duality routing plus abstention keeps the filter disengaged on benign hand scenes — safety by freezing versus safety while moving. On the static-obstacle suite the shields are competitive; the architecture gap is entirely where hazards are human or absent.

Against the unfiltered baseline: violations cut 12:1 at TSR parity. Pooled over 750 correctly-paired no-GT episodes, violation discordants are 12:1 in our favor (exact $p = 0.0034$) at TSR parity on oa/hs/aff ($p = 0.32/0.89/1.0$; aff parity reconfirmed at n=450, $p = 0.46$, surviving its winner's-curse audit). The GT tier goes further on obstacle avoidance: 77.8 vs. 73.6 baseline TSR at n=450 ($p = 0.045$) — the filter *helps* task success while filtering.

Honest disclosure: the oah freeze tax. On the dynamic-intruder suite our TSR is significantly below baseline (58.9 vs. 69.3 no-GT, $p = 0.0045$; 63.2 GT, $p = 0.029$): close-range approach gating around a moving hand still over-triggers. The baseline's higher oah success is partly bought with violations (5.3% of its episodes vs. our 1.7–1.9%). We report this as the price of the guard rather than tune it away; liveness refinements are under pre-registered evaluation.

Position provenance, for free. The GT vs. no-GT rows double as the position-provenance ablation of the identification stack: full perception costs ≈2 macro TSR points (69.6→67.6)

TABLE IV

LIBERO-SAFETY FIVE-ARM BOARD, SAFETY-FINETUNED $\pi_{0.5}$ CHECKPOINT, PER-SUITE TSR / VIOLATION% (BENCHMARK CONTACT PREDICATES; SUITES: OBSTACLE AVOIDANCE, HUMAN SAFETY, OBSTACLE+HANDS, AFFORDANCE). CELLS ARE n=150; BASELINE AND OURS CELLS FEEDING SIGNIFICANCE CLAIMS POOL n=450–750 (BATCH-AWARE PAIRING); MACROS FOLLOW THE POOLED RECOMPUTATION.

Arm	oa	hs	oah	aff	macro
$\pi_{0.5}$ baseline	73.6/4.7	85.8/0	69.3/5.3	56.7/0	71.4/2.5
Post-hoc shield	75.3/2.7	37.3/0	46.0/2.0	51.3/0	52.5/1.2
In-denoise reimpl.	70.7/0.7	32.7/0	48.0/1.3	48.7/0	50.0/0.5
Ours (GT)	**77.8**/4.2	**78.7**/0	**63.2**/1.9	**58.7**/0	**69.6**/1.5
Ours (no-GT)	68.7/**0.7**	**86.9**/0	**58.9**/**1.7**	**56.0**/0	**67.6**/**0.6**

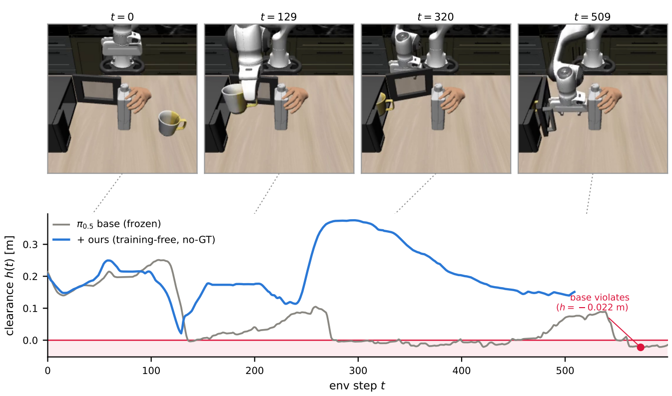

Fig. 5. The certificate at work over one episode (the pair of Fig. 1). Clearance $h(t)$ to the hand-attached hazard: the unfiltered base policy (gray) crosses h=0 at t=136 and spends 215 of 600 steps in violation; ours (blue) dips to +0.021 m at its closest approach and never crosses, completing the task at t=510.

and *reduces* violations (oa 4.2→0.7; the percep guard is conservative). This is the only ablation row of Sec. IX that requires no additional rollouts.

The refusal track. On the safety-reasoning suite (Table V) our judge refuses 100/100/80 of unsafe instructions with zero violations, vs. the benchmark's best published 80/72/36 — while the unguided policy *executes* them at 94/60/100. The execution-vs-refusal asymmetry is the knowing-doing figure: the policy hears language and complies; it does not judge.

IX. Experiments V: Which Literal Does the Work? Paired FOL Ablations

The novelty objection to any semantics-plus-CBF system is that the CBF machinery does all the work. We answer it causally: each arm deletes *exactly one* clause of Eq. (1) and re-runs the full no-GT stack, n=150/cell paired 1:1 against the incumbent on identical initial states, exact McNemar, incumbent column recomputed on the paired subset.[4] Figure 6 shows the deletions' guard sets on one scene; Table VI shows their cost at scale.

[4]Hence incumbent oah 56.0 here vs. 58.9 pooled in Table IV: the headline cells pool larger n; the ablation pairs against the matched subset.

TABLE V

SAFETY-REASONING (REFUSAL) TRACK: % OF UNSAFE INSTRUCTIONS REFUSED, BY LEVEL. THE JUDGE IS THE RULE-BASED SEMANTIC LAYER OF SEC. IV-D; THE UNGUIDED POLICY *executes* THE SAME INSTRUCTIONS.

	L0	L1	L2
Unguided policy (executes)	94	60	100
Ours (refuses)	**100**	**100**	**80**
Best published	80	72	36

TABLE VI

PAIRED SINGLE-CLAUSE DELETIONS, NO-GT TIER, n=150/CELL, EXACT MCNEMAR (DISCORDANT COUNTS IN PARENTHESES). FULL-METHOD COLUMN = PAIRED SUBSET OF THE INCUMBENT. §PENDING CELLS: OAH/NO-CLASS, AFF/GEOM-ONLY, AFF/NO-EXEMPT (QUEUED JOB).

Deleted clause	Suite	TSR full→abl.	Viol. %	p (TSR)
all logic (geom. top-k)	oa	68.7→44.0	0.7→**4.7**	1.2×10^{-6} (48:11)
	oah	56.0→39.3	1.3→0.7	2.5×10^{-4} (35:10)
	hs	88.7→68.7	0→0	5.3×10^{-6} (37:7)
¬MENT∧¬TGT	oa	68.7→40.0	0.7→1.3	1.8×10^{-8} (52:9)
	oah	56.0→39.3	1.3→0.7	6.2×10^{-4} (38:13)
PROT∨MOV	hs	88.7→82.0	0→0	0.099 (20:10)
	oah	pending§		
abstention (force-elect)	aff	56.0→42.7 (election chain, cached)		
VLM→hand-coded pred.	aff	56.0→47.3 (−9.6 at pooled n=450)		
GT→percep positions	all	= Table IV GT vs. no-GT rows		

Reading the matrix. Every completed arm confirms its pre-registered prediction. (1) **Geometry alone is not merely weaker — it is less safe**: the geom-only arm is the only deletion that *raises* violations (oa 0.7→4.7%, one-sided discordants 0:6, p = 0.031), because distance ranking both freezes the arm on false guards and evicts the true hazard from the top-k guard set. It also collapses TSR on every suite tested (−16.7 to −24.7). (2) **The language exemptions are worth ≈29 TSR points**: without ¬MENTIONED∧¬TARGET the filter guards the task's own target and self-paralyzes (oa 68.7→40.0). (3) **Abstention is worth ≈13 points** on the affordance suite: force-electing on benign scenes reproduces the known collapse (56.0→42.7 along the cached election chain), and swapping VLM predicate grounding for hand-coded tokens costs a further 9.6 — the VLM grounding is what scales. (4) **Class predicates**: on the static-hand suite the deletion trends negative but is not significant at n=150 (−6.7, p = 0.099) — consistent with hs having zero violations to prevent; the suite where PROTECTED∨MOVING should bind hardest (the dynamic intruder) is the pending cell, and until it lands we scope the class-predicate claim to a bounded trend. Deleting the state-aware heat gate changes no guard set on any cached affordance scene, so its rollout is identical by construction and it is reported offline-only.

X. Negative Results

Each mechanism below was implemented, pre-registered, and killed by its own gate; we report them because they bound the design space.

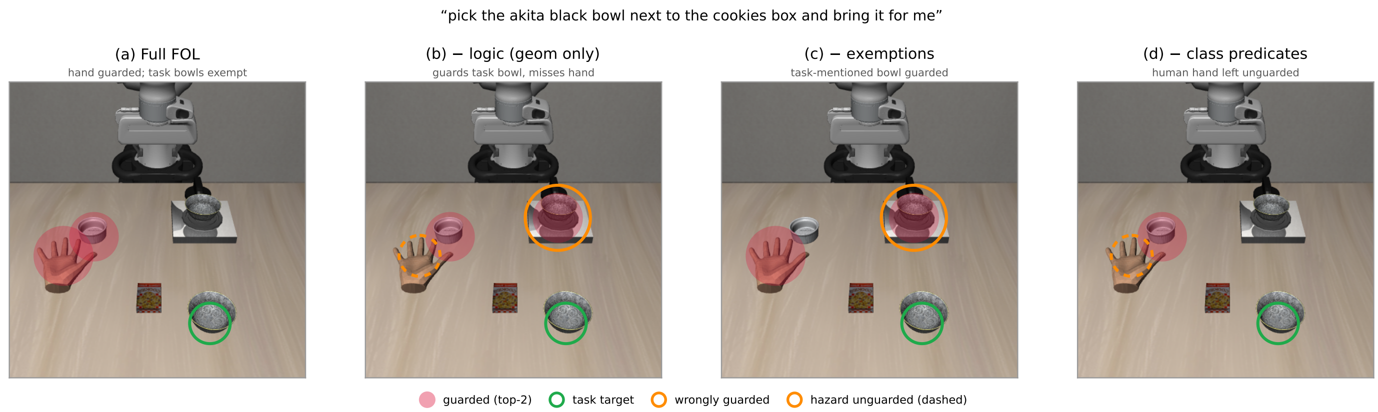

Fig. 6. What each conjunct of Eq. (1) buys, on one human-safety scene, computed by the deployed election code. (a) Full rule: the hand is guarded, the task's bowls are exempt. (b) Delete all logic (geometric top-k): the task's own bowl is guarded and the hand is missed. (c) Delete the language exemptions: a task-mentioned bowl is guarded — self-paralysis. (d) Delete the class predicates: the human hand is left unguarded.

Renoise repair (Sec. V): no gain over rejection sampling at equal compute, $p = 1.0$; NO-GO.

Safety prompting is not a safety mechanism. Across four testbeds — SafeLIBERO pilots (n=40: prompt-only 75.0/12.5 vs. baseline 67.0/14.0 — a TSR trend with CAR flat) and LIBERO-Safety L0 (n=50: obstacle prompt-baseline 20 TSR/8 violations vs. 12/4, violations *up*; human 46/0 vs. 54/0; prompting also hurts the guided arms) — the language channel moves task behavior, never avoidance. A fifth testbed closes the visual channel too: a $\pi_{0.7}$-style keep-out overlay drawn into the agentview input moves CAR not at all (12.5 vs. 14.0 baseline) while costing $\approx$8 TSR to occlusion. Conditioning is safety-blind in both modalities; enforcement must live in generation.

Scene graphs: NO-GO, three times. Graph-structured identification exactly ties the flat scorer wherever hazards are objects (SafeLIBERO 19/19 agreement; LIBERO-Safety obstacle 9/11 twice), and collapses identically on the relational human suite (0/2 top-2) — where the fix is a 10-line PROTECTED class rule (12/12), not structure.

VLM rule composition (FOL-compile). With *oracle* grounding the entire compositional apparatus gains at most one scene (via MOVING, adoptable as a flat class rule) and loses one to selection brittleness. With real VLM critics (two vendors) composition is strictly worse than the flat rule on every suite — haiku 13/16, gemini 11/13 vs. flat 16/19 on SafeLIBERO — and the characteristic failure is over-restrictive conjunctions that fire on *nothing*: the true hazard goes unguarded, fail-unsafe (2 and 10 empty-rule scenes respectively). Bounded on both sides, the direction is closed; its surviving artifacts are the two class rules and the predicate-grounding division of labor (Sec. IV-A).

Whole-scene VLM identification scores 18–64% (API-strength models) and 0–10% (local models, direct prompting) where the decomposed rule scores 100% on the same scenes — the decomposition claim survives at every model scale we tested.

Shape ablations. Superquadric/AABB barriers, corridor-relaxed shapes, and a faithful relation-conditioned "above" envelope all trade TSR for CAR without dominating the thin sphere + corridor; the semantic ideas that *did* transfer from the shape literature are the corridor exemption and the protected class, not the geometry.

XI. Limitations and Discussion

Honesty items. All SafeLIBERO comparisons vs. the SOTA method are against our faithful self-run reimplementation under one criterion (their published CAR is not reproducible under our criterion at any radius; see footnote to Table III); all LIBERO-Safety claims are against our self-run baseline on the released checkpoint (provenance footnote, Sec. VIII). LIBERO-Safety cell sizes are heterogeneous by design: every cell is at least n=150, and every cell feeding a significance claim was boosted to n=450–750 with batch-aware pairing; a uniform n=300 re-run of the full board is in progress and its two completed arms land within $\pm$3 pp of every frozen cell. The refusal-judge lexicon is in-domain (15 unique unsafe instructions); the quoted F1 is an upper bound pending held-out phrasings. The no-GT tier consumes the symbolic object *name* list; pixel-only naming currently costs identification (19$\rightarrow$12 of 19 top-2) and is future work.

Structural limitations. Obstacle-on-grasp cells resist every hard mechanism (the soft repulsor's 50% is the current ceiling); Long-horizon liveness stalls afflict all barrier-class methods including ours; the oah freeze tax is quantified and disclosed in the board rather than tuned away; identity errors fail unsafe at k=1 (mitigated by the guard-set mode); rotation is uncertified; benchmark-definition divergence (Long L2) shows a semantically-correct hazard model can disagree with a benchmark's arbitrary designated-obstacle label. *Part grounding*: the VLM layer already extracts part-level hazard semantics (blade vs. handle), but part *perception* failed our pre-registered side-discrimination gate (2/5 vs. the required 4/5), so all barriers are object-level — the semantic layer is currently ahead of what perception can ground, and a GT-part bracket (simulator part geometry) is the designed test of the available headroom.

Real-robot transfer. Three quantities decide transfer, and we state where each stands. *Sensing:* our margins are tuned to simulation-grade localization (median error 3.6–6.7 cm); real depth-camera error at close range is comparable, but the conservative-fallback path would engage more often, trading TSR for safety in the direction the GT/no-GT gap already measures. *Actuation:* A1$'$ absorbs tracking error into d_{safe} without a controller proof; an OSC tracking-error measurement on hardware would set d_{safe} empirically. *Timing:* SELECT/TILT add negligible latency at 20 Hz replanning; ASCEND's 57 ms per gradient step fits a 5-step replan budget at 1–2 steps; and the mover bound A2 ($\leq$8 cm/s at the deployed inflation) covers deliberate but not ballistic human hand motion — faster intruders need a larger λ or a predictive occupancy model in the PACS [3] style.

XII. CONCLUSION

Enforcement for frozen flow VLAs belongs in noise space. Hard repair — post-hoc or in-denoising — pays a paired, significant success tax and is never the safety winner; selecting, tilting, or ascending the noise keeps every action on the policy manifold and, composed with a soft velocity field, spans a certified, tunable success–safety frontier under one checked, mechanism-independent certificate. What to avoid needs no privileged state and no VLM authority: a three-literal rule with VLM-grounded predicates matches ground truth at zero cost, and every literal of the rule is causally necessary, measured by paired single-clause deletion. The sharpest system-level consequence is architectural: on both benchmark boards the dedicated shield architectures buy safety by freezing, while the assembled training-free stack keeps the policy moving — beating both shields by +15–20 macro TSR at equal-or-lower violations.

APPENDIX

APPENDIX A: PROOF SKETCHES

Thm. 1. With the brake u_j (approach components removed), $\|p_j + q - o_m\| \geq \|p_{j-1} + q - o_m\|$ for the argmin pair, so $B_j \geq B_{j-1} - \lambda$ under inflation growth; whenever $\gamma B_{j-1} \geq \lambda$ this implies the chain $B_j \geq (1-\gamma)B_{j-1}$. The push channel applies the correction along the radial escape direction of the argmin (obstacle, companion) pair, along which the effective distance strictly increases (for the sphere trivially; for the superquadric the shape term is constant along the center ray while distance grows); if the required push exceeds $[-c, c]^3$ the operator falls back to the full brake, which never decreases the barrier — the executed step is never a known-violating step. Below the λ/γ corner the push channel engages and Prop. 1 is what guarantees the executed prefix (checked, not assumed). *Assumptions:* single-integrator kinematics with linear command scaling (A1); tracking error absorbed by $d_{\mathrm{safe}} = 0.01$ m (A1$'$, no controller proof attempted); obstacle displacement $\leq \lambda$ per step within a chunk (A2); perception error shifts o_m and the certificate is relative to the estimate; the chain covers the translational executed prefix only (5 steps deployed), suffixes are re-planned.

Prop. 1. The acceptance stage rolls out $p_j = p_{j-1} + s\,u_j$ on the decoded chunk and evaluates the chain at full margins for $j \leq L$; passing is a predicate of the emitted actions alone. Failure of all K candidates triggers argmax-margin execution with the true margin reported.

Prop. 2. The lexicographic scalarization gives feasibility a dominating weight; ties among feasible candidates break by margin then progress. ■

APPENDIX B: DECLARED TASK-CLASS ROUTING PROTOCOL

Per-suite settings are a rule keyed on task class, declared once (per-task configuration is prohibited; it was ruled out as benchmark fitting in a pre-registered arm):

Task class (suite)	Identity	Gating
Geometric hazard (oa)	FOL	engage 0.22 m (+cone, GT)
Static human presence (hs)	FOLPROP	engage 0.30 m
Dynamic intruder (oah)	FOL	engage 0.30 m + cone 60°
Property hazard (aff)	FOLPROPVLM	engage 0.30 m

Shared by all cells: perception entity positions (no-GT tier), duality disengage, top-2 guard sets, execution parity, server ramp schedule with K=8 and end-effector radius 0.09 m. The class→configuration map mirrors the three-regime routing and is part of the method. Ablation arms (Sec. IX) reuse the incumbent configuration of their suite verbatim, changing only the election rule; full arm configurations and episode counts ship with the code.

REFERENCES

[1] Anonymous, "Neuro-symbolic safety guidance for vision-language-action models via constrained flow matching," arXiv:2607.01378, 2026.
[2] A. Agrawal and K. Sreenath, "Discrete control barrier functions for safety-critical control of discrete systems with application to bipedal robot navigation," in *Robotics: Science and Systems (RSS)*, 2017.
[3] Anonymous, "PACS: Provably-safe action chunking with failsafe trajectories for human-robot interaction," arXiv:2511.06385, 2025 (ICRA 2026).
[4] Anonymous, "VLS: Vision-language steering for frozen flow policies," vision-language-steering.github.io, 2026.
[5] Anonymous, "OmniGuide: Guidance injection for generalist robot policies," arXiv:2603.10052, 2026.
[6] Anonymous, "Constricting barrier functions for certified diffusion policy safety," arXiv:2602.21429, 2026.
[7] Anonymous, "Your model already knows: Reading VLA attention for obstacle-aware control," arXiv:2606.09749, 2026.
[8] W. Xiao et al., "SafeDiffuser: Safe planning with diffusion probabilistic models," arXiv:2306.00148, 2023.
[9] Anonymous, "AEGIS: A geometric safety shield for vision-language-action policies," arXiv:2512.11891, 2025.
[10] Anonymous, "COAST: Concept-level activation steering for vision-language-action models," arXiv:2605.17144, 2026.
[11] Anonymous, "LIBERO-Safety: Benchmarking safety-aware manipulation for vision-language-action models," arXiv:2606.23686, 2026.
[12] A. D. Ames et al., "Control barrier functions: Theory and applications," in *European Control Conference (ECC)*, 2019.
[13] L. Brunke et al., "Semantically safe robot manipulation: From semantic scene understanding to motion safeguards," *IEEE Robotics and Automation Letters*, 2025.
[14] Physical Intelligence, "$\pi_{0.5}$: A vision-language-action model with open-world generalization," arXiv:2504.16054, 2025.

[15] S. Liu et al., "Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection," in *ECCV*, 2024.